Deploy Geonetwork instructions

Goal-- deploy testing instance of GeoNetwork with DDE metadata configuration

requirements:
Geonetwork Github repository
https://opencode.deep-time.org/smrAzDDE/core-geonetwork/-/tree/DDEconfig
I have not been able to create a core-geonetwork repo in DDE-metadata

Apache Tomcat 9.0.85
Procedures were run on Windows 11, various things will be different under Linux or MacOS, but things should work-- its all Java, XML and JavaScript.

Goal:
Elasticsearch 8.11.3 running, with elastic user for whom you have the passwords
Apache Tomcat 9.0.85 running. The Geonetwork .war file is copied into the Webapps folder and Tomcat restarted to deploy Geonetwork
Geonetwork configuration must have the Elasticsearch 'elastic' user and password.
Geonetwork will be available at {localhost}:8080/geonetwork after Tomcat finishes deployment. for Demonstrations, the host server ('{localhost}') above must be publicly accessible on the web using its external IP address.

Steps.

1. get Apache Tomcat https://downloads.apache.org/tomcat/tomcat-9/v9.0.85/bin/ for your operating system. Instructions and background at https://tomcat.apache.org/tomcat-9.0-doc/setup.html
2. Get the DDE geonetwork.war file. This is a 'web' archive bundle that contains all the executable code, configuration files, and other resources needed for the application to work. Tagged releases will be made available from the DDE Gitlab repository tags (TBD, something like https://opencode.deep-time.org/{DDE}/core-geonetwork/-/tags), look for DDExxx tags. As of 2024-07-18, the current war file is at https://opencode.deep-time.org/dde/dde-medata/-/tree/geonetwork/GeoNetworkRelease, using GitLab large file storage.

3. The war file can be generated from the source code in the GitLab repository. This will require having a working version of Maven installed. Clone the GitLab repository that contains the source code, and build the .war from the source code, following the instructions at https://docs.geonetwork-opensource.org/latest/install-guide/installing-from-source-code/. NOTE the current source code is still at https://github.com/smrgeoinfo/core-geonetwork, I haven't figured out how to pull into GitLab.
4. Note, Geonetwork requires a Java 11 developer kit to be installed; I am using the https://adoptium.net/ version. Be sure to set your JAVA_HOME environmental variable to point to the JDK location e.g. (for Windows): '*C:\Program Files\Java\jdk-11.0.2*'
5. Install Elasticsearch 8.11.3 and Kibana 8.11.3. NOTE that you have to use equivalent versions for compatibility. Kibana is a front end tool for working with Elasticsearch, I'm not sure but it appears that Geonetwork uses some Kibana functionality. Both are installed by copying the contents of a zip archive to the location of your choice.
   a. Get elastic search at https://www.elastic.co/downloads/past-releases/elasticsearch-8-11-3
   b. Get Kibana at https://www.elastic.co/downloads/past-releases/kibana-8-11-3
6. start Elasticsearch, go to {elasticsearch home directory}/bin. With my Windows CLI console I need to type '.\elasticsearch' to get it going. Watch the stuff the spews out on your console, at the end there will be some important information. This is what you're looking for:

Γä╣∩╕Å Password for the elastic user (reset with `bin/elasticsearch-reset-password -u elastic`):
ApHUH7ZaQGB_J-cOeAVF
Γä╣∩╕Å HTTP CA certificate SHA-256 fingerprint:
62fe489812830713d49445d5fca607c997c7eaac2f5c6e6780b267560b97d82a
Γä╣∩╕Å Configure Kibana to use this cluster:
ΓÇó Run Kibana and click the configuration link in the terminal when Kibana starts.
ΓÇó Copy the following enrollment token and paste it into Kibana in your browser (valid for the next 30 minutes):

eyJ2ZXIiOiI4LjExLjMiLCJhZHIiOlsiMTAuMC4wLjE2Nzo5MjAwIl0sImZnciI6IjYyZmU0ODk4MTI4MzA3MTNkNDk0NDVkNWZjYTYwN2M5OTdjN2VhYWMyZjVjNmU2NzgwYjI2NzU2MGI5N2Q4MmEiLCJrZXkiOiJGSzM5VW80QjJxMUdJYldqVF8tZzpjU2dvam9xVVFmU0tkLWxoNE5BX3BnIn0=

I don't know if you'll see the same non- printing character stuff, but copy the elastic user pass word and the enrollment token. You can generate new ones if you need to but its easier to get these now. The enrolment token is only good for 30 minutes, so be ready to start Kibana as soon as you have it.

7. Start Kibana. same as Elastic search, go to {Kibana home directory}/bin and type './kibana. It takes several minutes to start the first time, then in the console it will give a URL to open a page that asks for the enrollement token. After it takes the token, it will ask for the 'elastic' user name and password. After providing that, kibana will open. that's it with Kibana for now
8. Next, you have to update some configurations. Quit Elasticsearch (control-C in the console window where you started Elasticsearch). ElasticSearch 8.11.3 automatically starts with security enabled. Getting it to work over https with Geonetwork requires setting up SSL certificates and other configurations; For development and testing, I haven't figured all this out, so have disabled the security on Elasticsearch. To do this open {elasticsearch home directory}/config/elasticsearch.yml edit the following keys:

```
xpack.security.enabled: false
xpack.security.enrollment.enabled: false
```

Its a simple key-value text file, so any text editor should work. I like Notepad++ with Windows. Save the configuration file.